DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 — TRYHACKME ROOM WRITEUPS

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Domain: Cybersecurity / SOC Analyst

Platform: TryHackMe

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ROOM 4: SNORT

Learning Objective

The objective of this room was to understand Snort, its operating modes, and how it can be used to monitor network traffic and detect suspicious activities.

1. Introduction

In this room, I learned about Snort, an open-source network intrusion detection and prevention system. It can analyse network traffic and use rules to identify suspicious packets or activities.

I understood that Snort is useful for security monitoring because it helps detect traffic that matches defined rules.

2. Key Concepts Learned

What Is Snort?

Snort is a network security tool used for packet analysis, intrusion detection, and intrusion prevention. It can inspect traffic and generate alerts when packets match configured rules.

Sniffing Mode

In sniffing mode, Snort reads network traffic and displays packet information. This helps understand communication between devices.

Packet Logging Mode

Snort can save captured network traffic to log files. Analysts can review these logs later to investigate network activity.

IDS Mode

An Intrusion Detection System (IDS) monitors network traffic and generates alerts when suspicious activity matches its rules.

IPS Mode

An Intrusion Prevention System (IPS) can detect suspicious traffic and take action to block it, depending on the configuration.

Snort Rules

Rules define which packets or activities Snort should detect. A rule can specify protocol, source and destination addresses, ports, flags, and alert messages.

PCAP Analysis

PCAP files contain captured network packets. Snort can read these files to analyse traffic and identify packets that match its rules.

3. Important Commands

```
# Check the Snort version
snort -V

# Test the configuration file
sudo snort -c /etc/snort/snort.conf -T

# Read a PCAP file
sudo snort -r capture.pcap

# Read a PCAP and display console alerts
sudo snort -c /etc/snort/snort.conf -q -r capture.pcap -A console

# Run Snort with full alert output
sudo snort -c /etc/snort/snort.conf -A full -l .
```

4. Practical Learning

During this room, I practised reading packet captures and understanding Snort's command-line parameters.

I learned how to investigate PCAP files, review packet statistics, and use rules to filter traffic based on packet information such as IP IDs, TCP flags, and UDP behaviour.

I also learned that the results depend on the PCAP file, the active rules, and the configuration being used.

5. SOC Analyst Perspective

Snort can help a SOC analyst identify suspicious network activity and investigate alerts. For example, a rule may detect unusual packet behaviour or traffic matching a known pattern.

The analyst can then review the related packets, source and destination addresses, ports, and timestamps to understand the activity.

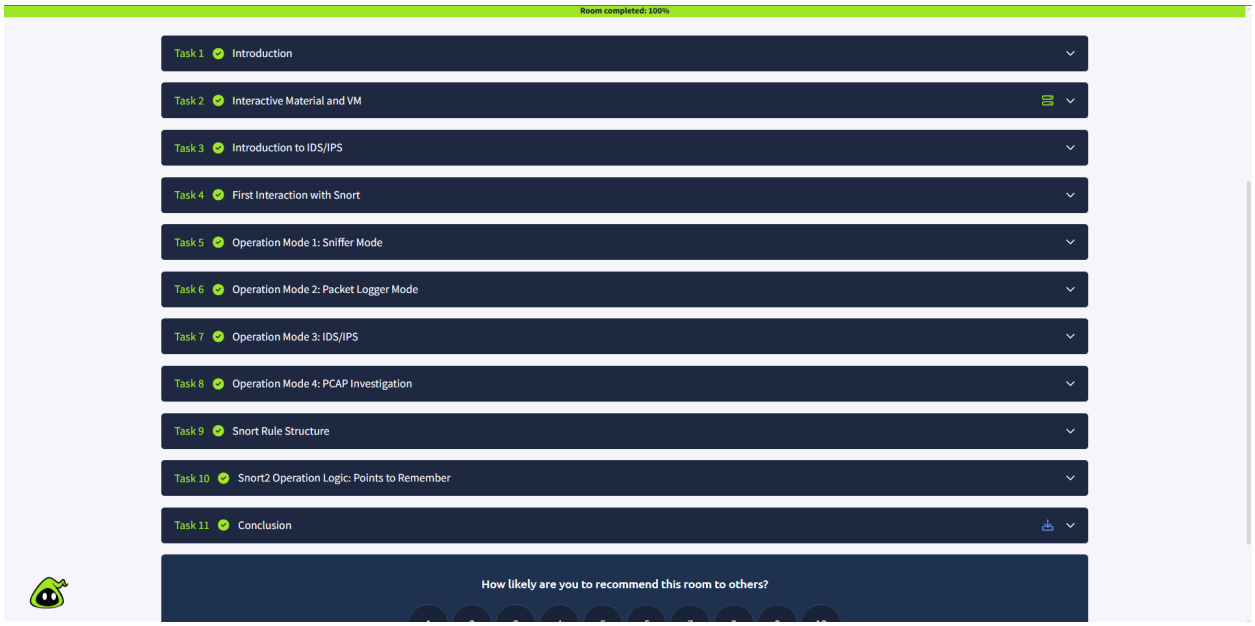
6. What I Learned

- I understood the purpose of Snort in network security.
- I learned the difference between IDS and IPS.
- I practised using Snort commands and configuration files.
- I learned how to investigate PCAP files.
- I understood how rules help detect specific packet patterns.
- I learned that alert results depend on the configured rules and captured traffic.

7. Conclusion

This room helped me understand how Snort can be used to monitor network traffic and investigate suspicious packets. I learned that packet analysis and rule-based detection are useful skills for SOC analysts.

8. Evidence


Room completed: 100%
Task 1 Introduction
Task 2 Interactive Material and VM
Task 3 Introduction to IDS/IPS
Task 4 First Interaction with Snort
Task 5 Operation Mode 1: Sniffer Mode
Task 6 Operation Mode 2: Packet Logger Mode
Task 7 Operation Mode 3: IDS/IPS
Task 8 Operation Mode 4: PCAP Investigation
Task 9 Snort Rule Structure
Task 10 Snort2 Operation Logic: Points to Remember
Task 11 Conclusion
How likely are you to recommend this room to others?